

Measuring the Value of World-Model Updates A Counterfactual Utility Protocol for Continual Adaptation

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Abstract

Continual world models must decide whether new data justify changing the model. Fixed replay schedules and prediction-error triggers specify when to update, but neither reveals the value of an individual update: one deployment run cannot show how the same model would have performed at that moment had it held its parameters. We introduce the *fork ledger*, which branches a deployment stream at pre-registered decision points into matched UPDATE and HOLD continuations under common random numbers. It evaluates both continuations on the same episodes and records $\Delta R = R_{\text{UPDATE}} - R_{\text{HOLD}}$. Always applying one fixed update mechanism lowers return on all three simulated control tasks: CartPole (-144.0 ; checkpoint-bootstrap 95% CI $[-185.4, -116.1]$, against a converged return near 650), Walker (-82.8 ; $[-101.1, -61.7]$) and Cheetah (-18.6 ; $[-29.0, -6.6]$). Divergence is an outcome of applying the update, so the estimand counts every attempted fork; restricted to the 693 of 720 that did not collapse, CartPole and Walker are unchanged in sign (-113.4 and -82.1) and Cheetah becomes unresolved (-3.9 ; $[-17.5, +13.0]$). The task is the unit of inference: each contributes 240 attempted forks over five pretrained checkpoints crossed with two drift directions. The ledger makes counterfactual utility observable for a fixed mechanism, allowing triggers to be judged by the updates they select rather than by surprise detection alone.

1 Introduction

Continual world models must decide when new data justify changing the model. DreamerV3 uses a fixed replay ratio; deployment-time adaptation can finetune a world model online, while model-management methods use model-shift, under- or overfitting, or change-point signals to decide when to update [Hafner et al., 2025, Feng et al., 2023, Ji et al., 2022, Dorka et al., 2023, Vovk et al., 2021]. These lines of work specify conditions for updating; the fork ledger asks whether an individual update improves control.

The missing quantity is counterfactual. Prediction error mixes reducible model mismatch with irreducible transition uncertainty, and an update can improve recent loss while degrading the representation used for control. At a decision point, the relevant comparison is therefore the return after updating versus the return the same deployed model would have obtained had it held its parameters. Existing model-management and retraining policies frame the decision around aggregate behavior, detection outcomes, or forecasted model quality rather than this matched per-update control contrast [Liebman et al., 2018, Regol et al., 2025, Steland, 2026].

The *fork ledger* makes the comparison observable in simulation. At each pre-registered decision point, it runs matched UPDATE and HOLD continuations under common random numbers and records $\Delta R_k = R_{\text{UPDATE},k} - R_{\text{HOLD},k}$ (Section 3.2). In the corrected sweep, always applying the one fixed

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update mechanism we study lowers return on CartPole and Walker, while Cheetah depends on drift direction and is unresolved. The stream-seed replication preserves this pattern. We therefore make a measurement claim and a task-bounded empirical claim, rather than propose a universal update rule.

Contributions.

- The fork ledger, a reusable matched counterfactual protocol that assigns a realized value to an individual update.
- Across three tasks, the independent unit of inference ($n_{\text{task}} = 3$, plus a legacy-schedule negative control), each with 240 attempted forks over five pretrained checkpoints and checkpoint-clustered intervals: one fixed rehearsal-free update mechanism is reliably harmful on all three, and its size changes with task and drift.
- A direct comparison of pre-fork trigger signals against the same labels. Residual and deployment-return triggers, the only families scored as decision policies in the corrected arm, are nearly tied on CartPole, where their ordering is largely a competence measurement, and neither ranking transfers across tasks.

2 Related Work

Detecting and timing updates. Novelty and out-of-distribution methods study when a learned model is unreliable [Zollicoffer et al., 2025, Nasvytis et al., 2024]. Event-triggered learning, concept-drift management, and retraining policies study when to adapt, assessing detection quality or aggregate downstream behavior [Ji et al., 2022, Dorka et al., 2023, Vovk et al., 2021, Liebman et al., 2018, Regol et al., 2025, Steland, 2026]. Off-policy evaluation can estimate the counterfactual, but only under assumptions about the response mechanism [Oberst and Sontag, 2019, Buesing et al., 2019]; in simulation, both branches can instead be run.

Residuals, adaptation, and update value. Likelihood and prediction-error signals can conflate model mismatch with data entropy or irreducible noise [Nalisnick et al., 2019, Caterini and Loaiza-Ganem, 2022, Zhang et al., 2021, Pathak et al., 2019, Mavor-Parker et al., 2022, Jarrett et al., 2023], and one-step likelihood need not track control performance [Lambert et al., 2020, You et al., 2026]. Recent systems use residual, consistency, or online-world-model signals for adaptation or evaluation [Domberg and Schildbach, 2026, Shi et al., 2026, Liu et al., 2025]. Closest is Ma et al. [2026], which couples noise across branches of a generative video model but intervenes on the action stream, where abduction is exact because the factual branch is self-generated; the fork ledger intervenes on the parameters, couples branches through simulator state and three randomness streams, and scores realized control return rather than plasticity loss or detection accuracy.

3 Method

3.1 Problem setting

Deployment stream. A world model p_θ is pretrained on a stationary environment and then deployed on a stream in which the environment changes. Two stream properties must be distinguished because a residual conflates them. **Volatility** is drift in a physics parameter over time (gravity for `cartpole`, contact friction for `walker`); an update can, in principle, track it. **Stochasticity** is Gaussian noise of scale σ_a added to the executed action; it raises transition uncertainty but is not an environmental change that an update can remove. We match drift profiles on accumulated level and rate so that residual detectability, rather than an unintended schedule difference, does not drive the comparison.

Control depends on the world model. Control uses latent-space MPC with cross-entropy-method (CEM) action-sequence optimization [Rubinstein, 1999], with a terminal value bootstrap in the style of TD-MPC2 [Hansen et al., 2024]. This choice matters: an amortised actor trained on the old latent space would confound a world-model update with actor mismatch, whereas MPC consults the current model at decision time, so an update changes behavior without actor retraining.

The update mechanism is held fixed. Every paired continuation applies N gradient steps to the most recent W transitions. We study *trigger times*, not competing update mechanisms; the labels

are consequently relative to this response. A 100-cell mechanism sensitivity changes the verdict only in the one task–direction combination where updating gains (Appendix A.4). The mechanism rehearses nothing: the deployment buffer is allocated fresh when the stream starts and the pretrained checkpoint stores model and optimiser state only, so the W -transition window holds post-deployment data exclusively. The ΔR labels therefore price no-rehearsal fine-tuning of a converged model, the regime in which catastrophic forgetting and unwanted drift are expected, not update value in general.

3.2 The fork ledger

The protocol has three stages (Figure 1).

1. Reference stream. A pretrained world model is deployed for $T = 6,000$ control steps ($K = 24$ decision points spaced 250 steps apart, at an action repeat of 4) under one cell: one pretrained checkpoint and one drift direction, at $\sigma_a = 0$ throughout the corrected arm. We log the one-step prediction error stream, the raw (unclipped) KL, the drift level $g(t)$ and rate $\Delta g(t)$, and the injected noise level. Freezing makes the same stream reusable for every trigger: an updated model’s error stream would depend on the trigger that produced it.

2. Paired labels. At K pre-registered decision points, we checkpoint and run two branches. UPDATE applies N gradient steps to the world model on the most recent W transitions, then evaluates. HOLD evaluates immediately. Both branches are evaluated on the same m episodes with identical initial states, injected action noise, and planner sampling noise. The physics perturbation is *pinned* at its value at the moment of the fork: return is measured under the world as it stands at the decision point, not one that keeps changing during evaluation. This choice has a cost. Pinning biases the contrast along the volatility axis, under-crediting updates where drift is faster if control loss is convex in staleness and *over*-crediting them if it saturates. Appendix A.6 quantifies both directions; which one obtains is empirical, not assumed, and the forward-window arm that bounds it empirically was run only on CartPole.

3. Scoring. We define ΔR as the causal effect of *applying the fixed update mechanism now versus not applying it*: it prices the entire intervention—timing, dose N , window W , and the induced model change together—and does not separate those components; it is not a timing effect with the dose partialled out. This is deliberate: it matches the decision a deployed trigger makes. Consequently, the base rate $\Pr(\Delta R > 0)$ carries a dose component, so the precision endpoint should be read against that base rate rather than against zero. A trigger is any function of the pre-fork error stream: at a chosen rate it partitions decision points into triggered and untriggered cases and is scored against the fixed ΔR labels, as pure post-processing. For the corrected signal comparison, the deployment-return feature is the mean return from the segment immediately before the fork, recorded before either branch is evaluated. We never use the post-fork R_{HOLD} column as a trigger feature. Common random

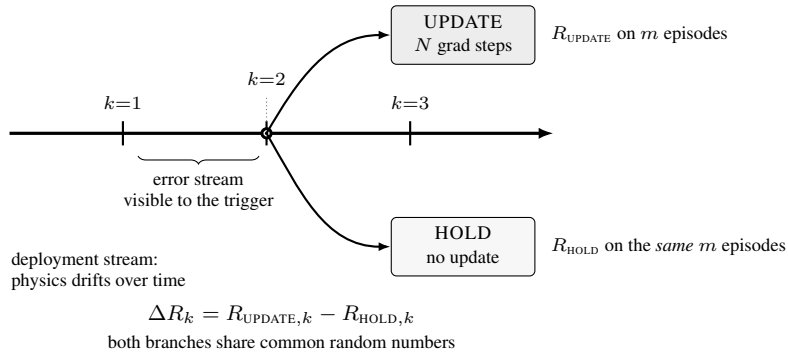


Figure 1: **The fork ledger.** A deployment stream is checkpointed at pre-registered decision points k and branched into UPDATE and HOLD continuations, evaluated on the *same* m episodes under common random numbers; ΔR_k is the realized value of updating at k . A trigger observes only the pre-fork error stream (brace), so every candidate is scored against the same labels without rerunning the environment.

numbers are standard practice [Glasserman and Yao, 1992]; three separate noise sources had to be pinned before repeated runs agreed.

3.3 Design corrections and pre-registered plan

Design and inference. *Corrected* cells run the committed protocol: a 20,000-step drift period, forks every 250 control steps, and a logged pre-fork deployment-return field. *Legacy* cells predate one or both; they supply sensitivity only, and those without the pre-fork field are quarantined from trigger ranking. The pre-registered design makes five choices explicit. Its primary endpoint, $\mathbb{E}[\Delta R \mid \text{triggered}] - \mathbb{E}[\Delta R]$ in units of the per-episode return standard deviation σ_{ep} , is not reported here, a deviation; mean cell-level ΔR , checkpoint-clustered AUC and policy totals stand in its place. A trigger reads only the pre-fork stream. AUC, whether a pre-fork scalar orders positive- ΔR forks above negative ones, is computed per checkpoint over its two pre-registered direction streams combined, and aggregated by checkpoint-cluster bootstrap rather than pooled over raw forks [Cameron et al., 2008]. All corrected cells run $\sigma_a = 0$ with $m = 30$ matched evaluation episodes per branch, the registered episode ramp not having been executed, so the stochasticity axis is unexercised. With only five clusters the percentile bootstrap under-covers; Student- t intervals on the same checkpoint means leave both negative verdicts intact (Appendix A.5), and no multiplicity adjustment is applied across the signal comparisons, which we read descriptively. We compare against a rate-matched random trigger, the ΔR oracle, and the registered drift oracle, which is simulator-only and so is a ceiling rather than a deployable rule. The registered design specified causal running-quantile thresholds; as a deviation from that registration, the corrected results instead fit the threshold level leave-one-checkpoint-out on the pre-fork signal. The rate-matched control permutes within each held-out cell, preserving the number of fires the reported policy made there; restricting it further to contiguous time blocks leaves the verdict unchanged wherever the restriction leaves the null any freedom (Section 4.5) [Anderson and ter Braak, 2003]. The registered choices and the deviations from them are recorded in Appendices A.1 and A.2.

Environment and model. The main study is proprioceptive cartpole-swingup from the DeepMind Control Suite [Tassa et al., 2018] on MuJoCo [Todorov et al., 2012], with walker-walk and cheetah-run replications and a legacy-schedule hopper-hop negative control (Section 4.1). The model is a compact DreamerV3-style recurrent state-space model [Hafner et al., 2025], trained from scratch on clean dynamics. The trainer reads back the constructed-environment flags to verify no perturbation or action noise was active during pretraining. The corrected grid and the registered trigger families are listed in Appendices A.1 and A.2.

4 Results

We report ΔR continuously rather than binarizing it into a precision label, which would attenuate the effect of noisy fork outcomes; the primary protocol and robustness checks are in Appendix A.

4.1 A fixed update rule loses return on all three tasks

The ledger turns an update rule into a policy over forks, allowing policies to be compared in return rather than detection rate (Figure 2). In the corrected CartPole sweep the mean cell-level effect over all 240 attempted forks is -144.0 $[-185.4, -116.1]$ against a converged return near 650. Excluding diverged forks conditions on a post-treatment variable, so the all-attempted mean is the estimand for update value and the retained mean answers the narrower question of what a non-collapsing update is worth. Under the divergence rule, which drops any fork whose update branch scores below 30% of its cell’s frozen-branch reference, 228 remain, updating helps on 58 and hurts on 170, the mean is -113.4 $[-131.2, -90.1]$, and all ten cells are negative. The harm is largest where drift is smallest: mean ΔR in the lowest drift quartile is -208.6 toward and -253.4 away, against -8.4 and $+53.0$ in the highest. The two directions traverse the drift range in opposite time order, so this is not stream position; the measured cost is not a failure to track drift.

The additional tasks bound the claim (Table 1). On Walker the two estimands agree, -82.8 $[-101.1, -61.7]$ over all attempted forks and -82.1 $[-100.2, -60.9]$ retained, with all ten cells negative under both drift directions. Cheetah is where they part: all attempted gives -18.6 $[-29.0, -6.6]$, reliably negative, while the retained subset spans zero (-3.9 $[-17.5, +13.0]$) with its sign set by the

direction the world moved, -26.8 toward the pretraining dynamics and $+19.0$ away. The rule biases ΔR toward zero, so the retained verdict is the conservative one.

A fourth task tests the instrument rather than the main claim. On hopper-hop, the legacy-schedule control, the planner never leaves the floor, so an update has little to change. The ledger reports $+1.0$ $[-0.3, +2.4]$ against the registered bound $|\Delta R| < 2.2$; the point estimate is inside the bound, although the upper endpoint is not. One caveat scopes this control: Hopper ran under the legacy drift schedule (period 2,000 steps, saturating from the first fork in the *away* direction) rather than the corrected 20,000-step schedule, so its drift-direction split is not comparable to the rows above; within that schedule, the control gives no evidence of a systematic negative shift.

The divergence rule (12 CartPole, 2 Walker, 13 Cheetah forks) is applied before any trigger is scored, and it conditions every reported effect on the forks it retains (Section 5). A separate legacy 46-cell CartPole dose sweep supplies endpoint and mechanism sensitivity (Appendix A.4).

4.2 Independent stream seeds preserve the task pattern

Two additional stream seeds per checkpoint and drift direction add 60 cells (30 per task combined with the original ledgers). The sign pattern is unchanged: CartPole and Walker remain negative in every cell, Cheetah unresolved (Table 2). The replication supports only the claim that those negative effects are not artifacts of a single stream realization; it does not turn three task families into nine independent tasks.

4.3 Why the update’s value varies: a registered mechanism, falsified

The CartPole result is not uniform across decision points. In that dose sweep, regressing R_{UPDATE} on R_{HOLD} across 1048 retained forks gives a slope of 0.27, not 1: the updated branch retains a quarter of

Table 1: All four tasks: ten cells and 240 attempted forks each; Forks counts those retained by the divergence rule, from which Hopper is exempt (its floor-level return would make the rule ill-defined). Returns are task-specific, never pooled; we average ΔR within cells and report checkpoint-cluster bootstrap intervals. Drift columns: *toward* decays the perturbation back toward the pretraining physics; *away* ramps it away from them. CartPole, Walker and Cheetah use the corrected pre-fork sweep; Hopper ran the legacy saturating drift schedule (see text) as the registered negative control.

Task	Cells	Forks (retained)	mean ΔR [95% CI]	mean ΔR by drift	
				toward	away
cartpole-swingup	10	228	-113.4 $[-131.2, -90.1]$	-152.6	-74.1
walker-walk	10	238	-82.1 $[-100.2, -60.9]$	-78.9	-85.3
cheetah-run	10	227	-3.9 $[-17.5, +13.0]$	-26.8	$+19.0$
hopper-hop	10	240	$+1.0$ $[-0.3, +2.4]$	$+0.2$	$+1.8$

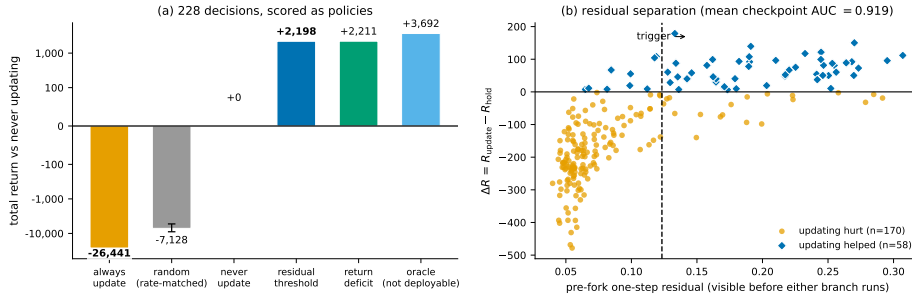


Figure 2: **Update rules as policies on the corrected CartPole sweep.** (a) Total return over 228 non-diverged forks at $N = 200$, relative to never updating (zero line), on a symmetric-log axis (linear within ± 100); the rate-matched random rule updates 70 times and shows its 95% permutation interval. (b) Each point is one fork; points right of the dashed median leave-one-checkpoint-out threshold trigger an update. The signal enriches helpful updates but does not separate them perfectly.

the variation in frozen performance, with standard deviation 76 versus 150. The fitted line crosses the identity at $R_{\text{HOLD}} = 397$. Binned mean ΔR is +55 below 350, +7 in $[350, 450)$, then -73 , -149 , and -240 as the frozen model improves (Figure 3), so most observed decisions lie in the region where this update is harmful.

Before running either additional task, we registered the following prediction: because Walker (297.2) and Cheetah (278) converge below CartPole’s crossing at 397, both should show ΔR between +70 and +110, with a pre-registered falsification threshold of $\Delta R < -50$ on either task. The corrected sweep gives Walker -82.1 , violating that criterion, and Cheetah -3.9 . Under the registered criterion the attractor account is therefore refuted as a transferable mechanism, and everything in this subsection past that verdict is exploratory. The per-task crossing analysis that follows is *post hoc*: fitting each task separately explains why the prediction failed. The crossing is a property of the *task*, not a transferable constant, but this analysis was not registered.

We also cannot claim the contraction generalises. The per-task legacy slopes, fit before the divergence exclusion, are 0.28 and 0.34, close to CartPole’s 0.27, but their cell-clustered intervals are $[-0.16, 0.76]$ and $[-0.43, 0.78]$. More decisively, a contraction toward a point must *shrink* the spread of outcomes, whereas the variance ratio is 3.14 $[2.05, 5.14]$ on Walker, evidence of expansion, and 1.35 $[0.71, 3.97]$ on Cheetah. The slopes agree because CartPole’s competence range is much wider than Walker’s, not because a common operator is at work.

Across tasks, the crossing itself does not transfer. Walker is negative under both drift directions, whereas Cheetah changes sign with the direction (Table 1), so the same update has task- and drift-dependent value even with the fork protocol held fixed. Measurement error and continual-learning pathologies such as loss of plasticity and primacy bias are two alternative explanations that we test [Dohare et al., 2024, Nikishin et al., 2022]; neither explains the observed task- and direction-dependent pattern (Appendix A.5).

The effect survives two objections to a single-cell result. Across that sweep’s $16 \times$ dose range over five independently pretrained checkpoints, mean ΔR is negative at every dose, and averaging within checkpoint leaves all five doses excluding zero (at $N = 400$, where four of the five checkpoints have cells, CI $[-159.0, -17.0]$, minimum detectable effect 89.8; Appendix A.5). The corrected multi-task sweep shows the sign is not an artifact of that dose grid, with Cheetah the boundary of the claim.

Table 2: Original corrected ledgers and independent stream-seed replication. Intervals are checkpoint-clustered. The replication adds stream variation within each task; it does not change the task-level uncertainty unit.

Task	original 10 cells	combined 30 cells
CartPole	$-113.4 [-131.2, -90.1]$	$-113.0 [-125.9, -97.4]$
Walker	$-82.1 [-100.2, -60.9]$	$-81.1 [-98.4, -62.2]$
Cheetah	$-3.9 [-17.5, +13.0]$	$-3.3 [-16.9, +10.8]$

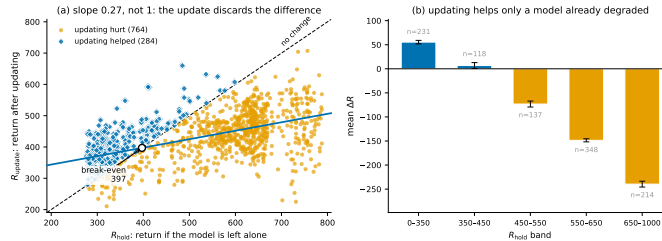


Figure 3: **Update value versus frozen-branch return in the legacy CartPole dose sweep (exploratory; the registered transfer prediction built on this fit was falsified).** (a) Each point is a retained fork across doses and checkpoints (1048 of 1088); the fit has slope 0.27 and crosses the identity at $R_{\text{HOLD}} = 397$. (b) Binned means show the sign change: updating helps when the model is already degraded, costs more as the frozen branch improves. Error bars are standard errors.

4.4 No fixed-rate schedule beats never updating on CartPole

A rule blind to task, drift direction and current competence, the quantities the sections above show ΔR tracks, cannot distinguish helpful from harmful updates, and neither family in use reads them. That dose sweep is an analogue of DreamerV3’s replay-ratio knob [Hafner et al., 2025]: at N gradient steps every 250 control steps, the effective ratio $N/250$ runs from 0.10 to 1.60. The analogy is limited: each arm sums independent one-shot fork contrasts on CartPole—a single N -step update of the frozen reference-stream model on its most recent W transitions—not a compounded replay schedule with full-buffer replay, and the arms are unequally sized (159 forks at the top of the range against 216 at the bottom, with legacy and corrected CartPole ledgers mixed at $N = 200$), so the family totals bound the range rather than rank it. Every setting of the ratio loses return against never updating, but per fork the loss runs from -82.7 to -114.1 with no monotone trend in N , and the decline in loss per gradient step across the range is the denominator growing rather than the update improving. No arm compounds, and none applies the periodic resets with which the replay-ratio literature buys higher ratios [D’Oro et al., 2023]. The corrected CartPole sweep gives the matched policy totals in Figure 2, at 200 gradient steps per update: both pre-fork signals recover positive utility, but neither reaches the oracle and this does not establish a universal trigger; of the registered families, only residual and random appear here (Section 4.5 accounts for those absent from the corrected comparison). Periodic retraining is likewise monotone toward zero in the legacy CartPole schedule baselines. All schedule evidence in this subsection is CartPole-only, from per-decision one-shot contrasts. This is a specification failure rather than a tuning failure: state-conditioned timing is useful on corrected CartPole, but the cross-task results show its learned ordering is not automatically portable. Learned replay scheduling can beat fixed schedules in continual learning [Klasson et al., 2023]; the ledger adds the realized per-decision value.

4.5 Trigger rankings do not demonstrably transfer across tasks

The corrected sweep makes the comparison possible without leakage: we rank the pre-fork residual and the deficit in mean return from the preceding deployment segment under the same leave-one-checkpoint-out calibration and ΔR labels. The residual trigger was registered; the return-deficit feature was added after registration and is exploratory. The registered raw-KL signal, scored on the same ledgers and retention rule, tracks the residual (AUC 0.917/0.502/0.674 across the three tasks). The registered drift oracle, a simulator-only ceiling, ranks better than either deployable signal and earns less: AUC 0.937/0.409/0.701 against policy totals $+1596/-1948/+800$, which is the ranking-is-not-utility gap in one row. As a deviation from the registration, four of the eight registered families are missing from this comparison: the periodic schedule ran only on quarantined legacy ledgers, and running variance, percentile, and held-out validation loss were never run (Appendix A.2). On CartPole the two policy signals we run are effectively tied: AUC 0.919 [0.869, 0.966] versus 0.915 [0.870, 0.959], with policy totals $+2198$ and $+2211$ on 228 retained forks against an oracle of $+3692$ (Appendix A.3). A rate-matched random residual rule earns $-7128 [-8949, -5303]$ (20000 cell-stratified permutations), so the CartPole totals are not a generic consequence of triggering; this control was pre-specified and run on all three tasks. Restricting the same control to contiguous time blocks preserves the verdict wherever the restriction leaves the null any freedom: the null total is -523 at two blocks and $+1599$ at four, both $p < 0.001$ against the observed $+2198$, while at six blocks each block holds three to four forks and the null converges on the observed value by construction. The CartPole total is also direction-dependent: the residual rule earns $+2033$ *away* versus $+165$ *toward*, and the return-deficit rule $+1856$ versus $+355$, so the positive aggregate is not a claim of equal benefit across drift directions.

The ordering is not universal, and the control does not pass everywhere. On Walker only five of 238 retained forks have $\Delta R > 0$, so its AUCs are barely defined: three of the five checkpoint clusters contain no positive fork, and the Walker brackets in Table 3 are min–max over the two that remain, one of them resting on a single positive fork; we draw no ranking conclusion from them. Both learned policies stay harmful (-1010 and -681) and each lies inside its own rate-matched control ($-945 [-1151, -750]$ and $-618 [-764, -475]$), so both fail there the check CartPole passes; Walker’s oracle is $+21$, so no rule has headroom. On Cheetah the two signals swap (0.680 versus 0.566, totals $+925$ and $+1310$ against an oracle of $+3697$) and both controls are cleared, but conditionally: with the divergence rule off the totals fall to $+16$ and $+100$ and clear neither ($p = 0.41$, $p = 0.27$), because Cheetah’s 13 excluded forks sit at the 0.8 quantile of the retained residual distribution, where the trigger fires. CartPole’s 12 sit at 0.2 and its totals are unchanged, so CartPole is the only task that

passes this control unconditionally. The data support a protocol for testing triggers and a single-task demonstration that a pre-fork signal can select less damaging moments, not a universal rule and not a cross-task reordering.

A ranking score is not policy utility. The AUC asks only for that ordering within a deployment stream; a policy must also turn the ordering into a decision rule and pay the value of every selected update, so each threshold is fit on four checkpoints and scored on the fifth.

An ordering can also be a proxy for something else. On CartPole the residual’s correlation with ΔR falls from $+0.756$ to -0.125 once R_{HOLD} is held fixed, and both pre-fork signals rank almost inversely with it (Spearman -0.914 and -0.956), so what they order there is competence; on Cheetah the residual survives that conditioning ($+0.362$ raw, $+0.269$ partial). A ranking diagnostic can establish that a signal is useful for a particular deployment stream; it cannot by itself establish that the resulting policy is useful, portable, or optimal.

We also do not pool these totals: available oracle value is roughly $+3,700$ on CartPole and Cheetah but $+21$ on Walker, so an aggregate would follow wherever the simulator left headroom.

5 Discussion

The fork ledger supplies the check that aggregate-return comparisons omit: the realized value of applying one update at one decision point. In this study, the sign of update value varies with task and drift direction, quantities a fixed replay ratio does not read. The CartPole trigger comparison shows that a signal can select useful updates within a task; Walker and Cheetah show why that ordering should be tested rather than assumed to transfer. By design the instrument never runs a continually adapting model: the stream continues from the HOLD branch, so each ΔR prices one update against a never-updated reference. A committing variant, in which fired updates persist, is the natural successor instrument.

What the paired ledger isolates. On CartPole the rate-matched random rule and the residual trigger spend the same 14,000 gradient steps, yet one loses $-7,128$ return units and the other earns $+2,198$. With matched counterfactual branches, that difference measures whether the rule chose better moments for the same intervention, an inference a trajectory-level aggregate conceals.

Scope and disclosures. The ledger needs restorable simulator state, a scalar downstream outcome, and one fixed update mechanism. Proprioceptive control supplies all three cheaply; interactive video world models supply the first two and lack only an agreed downstream outcome. It is an evaluation instrument, not a deployable trigger policy; the registered drift oracle is scored here but is simulator-only, since a deployment cannot read its own drift level. Three tasks carry return dynamics; the task-level result is therefore that the sign varies across the studied tasks and drift directions, not a population estimate. The legacy-schedule Hopper control is exempt from the divergence rule because its floor-level return would make the rule ill-defined. The registered arms with no committed ledger are enumerated in Appendix A.2. The parallel-episode cost model was calibrated only to 30 episodes, below the 60 per fork our cells run.

Table 3: Pre-fork trigger comparison across the three corrected task sweeps, ten cells each. Both signals are recorded before either branch runs and use the same leave-one-checkpoint-out threshold rule; each random control is matched to its own policy’s firing pattern. AUCs are checkpoint-clustered; policy totals are task-specific and never pooled. Walker’s AUC brackets are min–max ranges over the only two checkpoint clusters containing a positive fork, not bootstrap intervals (Section 4.5).

Task	forks	residual AUC	return AUC	LOO ΔR (res. / ret.)	oracle	random control (res. / ret.)
cartpole	228	.919 [.869, .966]	.915 [.870, .959]	$+2198 / +2211$	$+3692$	$-7128 / -6889$
walker	238	.447 [.234, .659]	.540 [.506, .574]	$-1010 / -681$	$+21$	$-945 / -618$
cheetah	227	.680 [.524, .812]	.566 [.428, .670]	$+925 / +1310$	$+3697$	$-74 / +201$

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A Protocol and robustness notes

A.1 Primary protocol

The corrected primary arm uses one fixed update response: $N = 200$ gradient steps on the most recent $W = 10,000$ transitions, evaluated on $m = 30$ matched episodes per branch at $\sigma_a = 0$. The corrected grid per task is five pretrained checkpoints $\times$ two drift directions (10 cells and 24 attempted forks per cell) under a 20,000-step drift period. The model is pretrained only on clean dynamics; every paired continuation pins the initial state, injected-action-noise stream, and recurrent-model sampler. No trigger, threshold, or update setting is chosen from the ledger it later scores. The accompanying code artifact contains the harness, the analysis scripts behind the corrected results, and the corrected fork ledgers, whose JSON records the executed configuration of every cell.

A.2 Deviations from the registered design

The registered evaluation-episode ramp (30, 75, 120 across the three σ_a cells) was not executed: every committed corrected-arm ledger runs $m = 30$ at $\sigma_a = 0$, so the stochasticity axis of Section 3.1 is unexercised in the corrected arm (the quarantined legacy CartPole ledgers shipped for sensitivity include $\sigma_a \in \{0.3, 0.6, 1.2\}$ cells). Of the eight registered trigger families (thresholded one-step error, running variance, percentile, raw KL, held-out validation loss, periodic, random, and drift oracle), the corrected arm scores four—thresholded one-step error, raw KL (as a ranking signal, Section 4.5), the drift oracle, and rate-matched random triggering—alongside one unregistered addition, the deficit in

pre-fork deployment return, and the ΔR oracle ceiling. The periodic schedule has results only on the quarantined legacy ledgers; running variance, percentile, and held-out validation loss have no result anywhere. The registered causal running-quantile threshold rule was not used: corrected thresholds are fit leave-one-checkpoint-out on the pre-fork signal (Section 3.3). The registered primary endpoint is not reported (Section 3.3). Hopper’s negative control ran the legacy 2,000-step drift schedule rather than the corrected 20,000-step schedule (Section 4.1).

A.3 Policy and trigger totals

On the 228 retained CartPole forks, always updating totals $-26,441$ return units relative to never updating. Leave-one-checkpoint-out residual and return-deficit triggers total $+2,198$ and $+2,211$, respectively; the unachievable ΔR oracle totals $+3,692$. Across tasks, residual and return-deficit AUCs are nearly tied on CartPole (.919 and .915), are defined on only two of five checkpoint clusters on Walker (.447 and .540, min–max ranges rather than intervals), and overlap on Cheetah (.680 and .566). Returns are never pooled across tasks.

A.4 Endpoint and mechanism robustness

The corrected *CartPole* endpoint remains negative at three update doses: mean cell-level $\Delta R = -88.4$ ($[-102.0, -71.9]$) at $N = 50$, -113.4 ($[-131.2, -90.1]$) at $N = 200$, and -110.1 ($[-132.2, -86.6]$) at $N = 400$; the dose arm was not run on Walker or Cheetah, so dose insensitivity is established only for CartPole. A separate 100-cell update-mechanism sweep varying the window to $W = 2,000$ and $W = 50,000$ changes the magnitude of the harm and, on cheetah ascending, its sign, so the body’s labels are conditional on the fixed mechanism. The independent stream-seed replication preserves the negative CartPole and Walker effects while leaving Cheetah unresolved.

A.5 Measurement error, plasticity, and the dose interval

Every interval in the body is a percentile bootstrap over five checkpoint clusters, where the 2.5% quantile sits near the smallest cluster mean and coverage is optimistic. The Student- t intervals on the same five means are $[-146.7, -80.0]$ on CartPole, $[-111.7, -52.5]$ on Walker and $[-28.4, +20.6]$ on Cheetah, so the two negative sign verdicts hold under either choice and Cheetah stays unresolved under both. No multiplicity adjustment is applied across the signal comparisons, which we read descriptively. Splitting each fork’s frozen episodes into independent halves reproduces the identical noise structure with no update present and gives a null slope of 0.98 ($[0.98, 0.99]$) with a variance ratio of 1.00 , against 0.47 and 0.44 observed on the same fork population (all 1821 divergence-retained forks across the three tasks, pooled only for this noise diagnostic), so measurement error does not produce contraction of the kind in Section 4.3. On the legacy CartPole sweep, holding R_{HOLD} fixed, the partial correlation of ΔR with fork index is $+0.09$, and sixteen of seventeen comparable cells contain both signs of ΔR , which monotone cumulative plasticity loss cannot produce. The $N = 400$ dose interval $[-159.0, -17.0]$ averages over the four checkpoints with cells at that dose and uses $t_{.975,3} = 3.182$; an earlier “underpowered” reading used $t_{.975,2} = 4.303$ from a shifted degrees-of-freedom row.

A.6 Pinned-physics bias and the forward-window arm

In a toy staleness model the forward/pinned contrast ratio is 1.000 for linear control loss, rises to 1.98 – 10.8 for convex loss as drift increases, and reverses under saturating losses ($0.791 \rightarrow 0.209$), so curvature, not drift rate alone, sets the direction of the pinning bias. The completed forward-window arm bounds it empirically on CartPole: over ten matched cells, forward minus pinned mean cell-level ΔR is $+3.9$ (checkpoint-bootstrap 95% CI $[-2.4, +10.2]$), and the leave-one-checkpoint-out policy totals remain close ($+2042$ versus $+2198$ for the residual trigger, $+2210.6$ versus $+2211.4$ for the return deficit). The CartPole headline endpoint is therefore not explained by pinning the evaluation physics; this arm was not run on Walker or Cheetah, so it does not clear them, and the interval leaves small sensitivity unresolved. The larger analyses—the continuous-endpoint derivation, common-random-number audits, and the β sensitivity—do not fit in this workshop paper; they qualify the scope of the measurement but do not add a second headline result.